

AP BIOLOGY · THE UNIVERSAL STUDY GUIDE

Targeting a 3, 4, or 5 — every concept tagged so you know what your tier needs

How to read the tier badges: [3] foundational · [4] adds value for 4+ · [5] adds value only for 5

Reading rule: the badge marks the LOWEST score tier this matters for. Targeting a 4 → study [3] AND [4]. Targeting a 5 → study all three.

PAGE 1 | MATH OF 3/4/5 · 8 UNITS · THE UNIVERSAL BIO FRAMEWORK

THE MATH OF YOUR TARGET — what you actually need to score

Pick your target. The score breakdown below shows exactly what you need to bag in each section. None of these requires you to memorize every detail — they reward strategic preparation matched to your target.

Target Score	MCQ (60 q, 50%)	FRQs (8 q, 50%)	Approx % needed	What that takes
3 (PASS)	~32 / 60 (~53%)	Get points on every part	~50%	Foundational concepts, attempt every FRQ, bullet-point answers OK
4 (SOLID)	~38 / 60 (~63%)	1-2 mostly complete FRQs + partials on rest	~62%	Connect concepts across units, structured FRQ answers in context
5 (MASTERY)	~46 / 60 (~76%)	Substantially complete answers on most FRQs	~75%	Mechanism-level reasoning, sophisticated experimental design, scientific argumentation

Be honest with yourself about where you're starting from. The doc tags concepts so you can prioritize. If aiming for a 3, you can skip [5] sections entirely. If aiming for a 5, every tag matters.

THE 8 UNITS — by exam weight

#	Unit	Weight	Key topics
1	Chemistry of Life	8–11%	Water properties, macromolecules, pH, enzymes
2	Cell Structure & Function	10–13%	Organelles, membrane transport, prokaryotes vs eukaryotes
3	Cellular Energetics	12–16%	Photosynthesis, cellular respiration , ATP, enzyme kinetics
4	Cell Communication & Cell Cycle	10–15%	Signaling pathways, cell cycle, mitosis, cancer
5	Heredity	8–11%	Mendelian genetics, Punnett squares, sex-linked traits
6	Gene Expression & Regulation	12–16%	DNA replication , transcription, translation, mutations, operons
7	Natural Selection (Evolution)	13–20%	Hardy-Weinberg , speciation, phylogenetic trees
8	Ecology	10–15%	Population dynamics, energy flow, biogeochemical cycles

Units 3, 6, and 7 are 37–52% combined. Photosynthesis/respiration, gene expression, and evolution dominate the FRQs.

THE 4 BIG IDEAS — the framework College Board uses to organize all of bio

Code	Big Idea	What it captures
EVO	Evolution	Living systems evolve through interactions with their environment (Unit 7 + connects everywhere)
ENE	Energetics	Biological systems use energy and molecular building blocks (Unit 3 lives here)
IST	Information Storage and Transmission	Living systems store, retrieve, transmit, and respond to information (Units 4, 5, 6)
SYI	Systems Interactions	Biological systems interact, and these systems and their interactions exhibit complex properties (Units 2, 8)

KEY INSIGHT [5] Every FRQ rubric ties answers back to one of the 4 Big Ideas. Naming the Big Idea explicitly in your response — "This relates to **EVO** because populations evolve when..." — earns sophistication credit. Most 4-scorers don't do this; that's part of why they're not 5-scorers.

PAGE 2 | MCQ MASTERY + MASTER VOCABULARY

MCQ STRATEGY — 60 questions in 90 min (~90 sec/Q)

- [3] Read the entire question first, then ALL four answer choices. Most AP Bio MCQs have 2 plausible answers — pick the more specific/complete one.
- [3] Eliminate impossible answers first. Wrong direction (decreasing when problem says increasing); wrong magnitude (10× when answer is 2×); wrong process (mitosis when problem is meiosis).
- [3] Watch for compound questions. "Which of the following is true about X AND Y?" — both halves must be correct. One wrong half = eliminate.
- [4] Data-based MCQs are 30+ % of the section. Practice reading graphs/tables/diagrams quickly. Identify the variables first, THEN answer.
- [4] When given experimental data, look for: control vs experimental groups, sample sizes, error bars (significant overlap = no clear difference).
- [5] Use answer-choice patterns. When 3 of 4 answers share an attribute and the 4th is different, the different one often signals a deliberate trap (or the answer).

MASTER VOCABULARY — 30 high-frequency terms

Term	Definition + tier
Hydrogen bonding [3]	Weak attraction between H of one water + lone pair of another. Source of water's high specific heat, cohesion, surface tension.
Hydrophilic / hydrophobic [3]	Polar / nonpolar. Drives membrane structure (phospholipid bilayer).
Enzyme [3]	Protein catalyst. Lowers activation energy. Active site specificity. Affected by temperature, pH, [substrate].
Substrate [3]	Molecule an enzyme acts on. Fits in active site (induced fit model).
ATP [3]	Adenosine triphosphate. Energy currency. Releases energy when hydrolyzed to ADP + P _i .
Eukaryote / prokaryote [3]	Has / lacks membrane-bound organelles + true nucleus. Eu = animals, plants, fungi, protists. Pro = bacteria, archaea.
Phospholipid bilayer [3]	Two layers of lipids with hydrophilic heads out, hydrophobic tails in. Selectively permeable.
Active vs passive transport [3]	Passive: down concentration gradient (no ATP). Active: against gradient (ATP-powered, e.g., Na/K pump).
Osmosis [3]	Diffusion of WATER across membrane. Hyper/hypo/isotonic — water moves toward HIGHER solute concentration.
Diffusion [3]	Net movement of molecules from high to low concentration. Faster for small/uncharged molecules.
Mitochondria / chloroplast [3]	ATP factories (mito = respiration). Photosynthesis factories (chloro). Both have double membranes (endosymbiotic origin).
Glycolysis [3]	Glucose → 2 pyruvate. In cytoplasm. Net: 2 ATP, 2 NADH.
Krebs (citric acid) cycle [3]	Pyruvate fully oxidized. In mitochondrial matrix. Per glucose: 2 ATP, 6 NADH, 2 FADH ₂ .
Electron transport chain (ETC) [3]	Inner mito membrane. NADH/FADH ₂ donate e ⁻ ; H ⁺ pumped into intermembrane space. ATP synthase makes ~28 ATP.
Light reactions [3]	Thylakoid membrane of chloroplast. H ₂ O split, NADPH and ATP made. O ₂ released.
Calvin cycle [3]	Stroma of chloroplast. CO ₂ + NADPH + ATP → G3P → glucose. Light-INDEPENDENT but needs light products.
Mitosis vs meiosis [3]	Mitosis: 1 division, 2 identical daughter cells (somatic cells). Meiosis: 2 divisions, 4 genetically unique haploid gametes.
Crossing over [3]	Homologous chromosomes exchange DNA in meiosis prophase I. Source of genetic variation.
Independent assortment [3]	Homologous chromosome pairs separate independently. 2 ⁿ possible combinations (n = haploid number).
Allele [3]	Variant of a gene. Dominant (capital) vs recessive (lowercase).
Genotype / phenotype [3]	Genetic makeup (e.g., Aa) vs observable trait (e.g., "tall").
Codominance vs incomplete dominance [4]	Codominance: both alleles fully expressed (AB blood). Incomplete: blended phenotype (red × white = pink).
Sex-linked [4]	Gene on X (or Y). X-linked recessive disorders affect males more often (X ^y Y pattern).
Transcription [3]	DNA → mRNA in nucleus. RNA polymerase reads 3'→5', synthesizes 5'→3'.
Translation [3]	mRNA → protein at ribosome. Codons (3 bases) → amino acids via tRNA.
Mutation [3]	Change in DNA sequence. Point: substitution. Frameshift: insertion/deletion shifts reading frame (more harmful).
Operon (lac, trp) [4]	Cluster of genes regulated together in prokaryotes. Lac is INDUCIBLE (turned ON by lactose); trp is REPRESSIBLE (turned OFF by tryptophan).
Hardy-Weinberg equilibrium [4]	$p^2 + 2pq + q^2 = 1$. Gives expected allele frequencies in non-evolving population. Deviation = evolution.
Phylogenetic tree [3]	Branching diagram of evolutionary relationships. Nodes = common ancestors. Branch length sometimes represents time.
Trophic level [3]	Position in food chain. Energy decreases ~10% per trophic level moving up.

UNIT 1 | CHEMISTRY OF LIFE · water, macromolecules, enzymes

Big picture. All life is built from carbon-based macromolecules in water. Water's unique properties (cohesion, high specific heat, polarity) make life possible. Enzymes regulate every reaction.

Water's life-supporting properties [3]

- **Polar covalent bonds + hydrogen bonding** between water molecules → high specific heat, surface tension, cohesion (water sticks to itself), adhesion (sticks to other surfaces — capillary action in plants).
- **Solvent of life** — dissolves polar/ionic molecules but not nonpolar (hydrophobic effect drives membrane formation).
- **Less dense as ice** — ice floats; insulates lakes in winter so life persists below.

The 4 Macromolecules [3]

Type	Monomer	Examples + function	Key features
Carbohydrates	Monosaccharide (glucose, fructose)	Glucose (energy), starch (plant storage), glycogen (animal storage), cellulose (plant cell wall)	Quick energy + structure. Glycosidic bonds.
Lipids	Glycerol + fatty acids	Triglycerides (fat storage), phospholipids (membranes), steroids (hormones)	Hydrophobic. Saturated (animal fats, solid) vs unsaturated (plant oils, liquid).
Proteins	Amino acid (20 types)	Enzymes, transport (hemoglobin), structure (collagen), antibodies, signals (hormones)	Peptide bonds. 4 levels of structure (primary–quaternary).
Nucleic acids	Nucleotide (sugar + phosphate + base)	DNA (genetic info), RNA (translation, regulation)	Phosphodiester bonds. Antiparallel strands in DNA.

Protein Structure — 4 levels [4]

- **Primary** — amino acid sequence determined by gene. Linear order of AAs.
- **Secondary** — α -helix or β -sheet from H-bonds between backbone (NOT side chains).
- **Tertiary** — 3D folding from interactions of R-groups (side chains): H-bonds, ionic, hydrophobic, disulfide bridges.
- **Quaternary** — multiple polypeptide subunits assembled (e.g., hemoglobin = 4 subunits).

KEY INSIGHT [5] Denaturation is when proteins lose their 3D shape (and thus function) due to heat, pH, salt. This is why fevers are dangerous — enzymes denature above $\sim 104^{\circ}\text{F}$.

Enzymes — biological catalysts [3]/[4]

Enzymes are proteins that lower activation energy without being consumed. Each enzyme has a specific active site that binds its substrate (induced fit model).

Factors that AFFECT enzyme activity (and how):

- Temperature — increases rate UNTIL denaturation (typically $\sim 37^{\circ}\text{C}$ optimum for human enzymes)
- pH — each enzyme has optimal pH (e.g., pepsin ~ 2 in stomach; trypsin ~ 8 in intestine)
- [substrate] — increases rate until enzyme saturated (V_{max})
- Inhibitors:
 - COMPETITIVE: binds active site (overcome with more substrate)
 - NON-COMPETITIVE/ALLOSTERIC: binds elsewhere, changes shape (NOT overcome by more substrate)

[4] Allosteric regulation: molecules bind a regulatory site (NOT the active site), shifting protein between active and inactive forms. **Feedback inhibition** is when the END product of a pathway inhibits an early enzyme — efficient regulation.

Acids, Bases, Buffers [3]

- **pH** = $-\log[\text{H}^+]$. pH 7 = neutral. Each unit = $10\times$ difference.
- **Buffers** maintain pH by absorbing or releasing H^+ . Critical for life — most enzymes function in narrow pH range.

UNIT 2 | CELL STRUCTURE & FUNCTION

Prokaryotic vs Eukaryotic Cells [3]

Feature	Prokaryotic (bacteria, archaea)	Eukaryotic (plants, animals, fungi, protists)
Nucleus	No (DNA in cytoplasm = nucleoid)	Yes (DNA in membrane-bound nucleus)
Organelles	No membrane-bound organelles	Many (mitochondria, ER, Golgi, lysosomes, etc.)
Size	Small (~1–10 μm)	Larger (~10–100 μm)
DNA	Circular, no histones	Linear, packaged with histones into chromatin
Ribosomes	70S	80S (cytoplasm); 70S in mitochondria/chloroplasts
Reproduction	Binary fission	Mitosis + meiosis

KEY INSIGHT [4] Endosymbiotic theory: mitochondria + chloroplasts originated as engulfed prokaryotes. **Evidence:** double membrane, own circular DNA, 70S ribosomes, divide independently of cell.

Key Eukaryotic Organelles — function + the AP-relevant connections [3]/[4]

Organelle	Function + tier-relevant detail
Nucleus	Houses DNA. Controls gene expression. Nuclear envelope has pores for RNA/protein traffic. [4] Nucleolus inside makes ribosomes.
Mitochondria [3]	ATP production via cellular respiration. Double membrane (cristae increase surface area). Own DNA + ribosomes.
Chloroplasts [3]	Photosynthesis (in plants/algae). Double membrane. Thylakoid stacks (grana) inside; surrounded by stroma.
Endoplasmic Reticulum (ER) [3]	Rough ER (with ribosomes) — protein synthesis + processing. Smooth ER — lipid synthesis, detox, Ca^{2+} storage.
Golgi apparatus [3]	Modifies, sorts, packages proteins/lipids. "Post office" of cell.
Lysosomes [3]	Digestive enzymes (animal cells). Break down waste, old organelles. pH ~5 inside.
Vacuoles [3]	Storage. Plant cells: large central vacuole (water + solute storage, turgor pressure).
Ribosomes [3]	Protein synthesis. Free in cytoplasm (intracellular proteins) or on rough ER (secreted/membrane proteins).
Cytoskeleton [4]	Microtubules + microfilaments + intermediate filaments. Shape, movement, transport.
Cell wall [3]	Plants: cellulose. Fungi: chitin. Bacteria: peptidoglycan. Provides structure/protection.

Membrane Transport — the gateway [3]/[4]

PASSIVE (no ATP, down concentration gradient):

- Simple diffusion — small, nonpolar (O_2 , CO_2) cross directly
- Facilitated diffusion — through channel/carrier protein (water via aquaporin; glucose via GLUT)
- Osmosis — diffusion of WATER toward higher solute concentration

ACTIVE (requires ATP, against gradient):

- Sodium-potassium pump — pumps 3 Na^+ out + 2 K^+ in per ATP
- Endocytosis / Exocytosis — vesicle-mediated for large molecules

TONICITY (effect on cell):

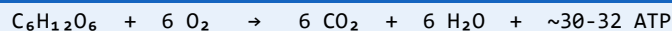
- Hypertonic — cell shrinks (water leaves)
- Hypotonic — cell swells / bursts (water enters)
- Isotonic — no net movement

KEY INSIGHT [4] Why surface area matters. Small cells have favorable SA:V ratio for transport. Large cells use organelles (ER, mitochondria) and folded membranes (microvilli, cristae) to maintain effective surface area.

UNIT 3 | CELLULAR ENERGETICS · photosynthesis + respiration

Big picture. Photosynthesis captures light energy and stores it in glucose ($\text{CO}_2 \rightarrow \text{C}_6\text{H}_{12}\text{O}_6$). Cellular respiration unlocks that energy from glucose to make ATP ($\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow \text{CO}_2 + \text{ATP}$). They're complementary — one's products are the other's reactants.

Cellular Respiration — 4 stages [3]/[4]



Stage 1: GLYCOLYSIS (cytoplasm)

Glucose $\rightarrow$ 2 pyruvate. Net: 2 ATP, 2 NADH. No O_2 needed.

Stage 2: PYRUVATE OXIDATION (mito matrix)

2 pyruvate $\rightarrow$ 2 acetyl-CoA + 2 CO_2 + 2 NADH

Stage 3: KREBS CYCLE (citric acid cycle, mito matrix)

2 acetyl-CoA $\rightarrow$ 4 CO_2 + 6 NADH + 2 FADH_2 + 2 ATP (per glucose)

Stage 4: ELECTRON TRANSPORT CHAIN + CHEMIOSMOSIS (inner mito membrane)

NADH/ FADH_2 donate e^- $\rightarrow$ pump H^+ out $\rightarrow$ ATP synthase uses H^+ flow back in to make ATP

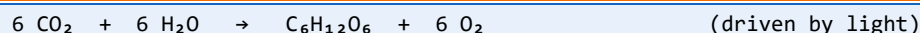
O_2 is FINAL e^- acceptor $\rightarrow$ forms H_2O . Yields $\sim 26\text{-}28$ ATP.

Anaerobic Respiration / Fermentation [3]

Without O_2 , cells can't run ETC $\rightarrow$ NADH accumulates $\rightarrow$ glycolysis stalls. **Fermentation regenerates NAD^+** so glycolysis can continue, producing only 2 ATP per glucose.

- **Lactic acid fermentation** (animals, some bacteria): pyruvate $\rightarrow$ lactic acid. Burns muscles during anaerobic exercise.
- **Alcoholic fermentation** (yeast): pyruvate $\rightarrow$ CO_2 + ethanol. Beer, bread, wine.

Photosynthesis — 2 stages [3]/[4]



Stage 1: LIGHT REACTIONS (thylakoid membrane)

Light splits $\text{H}_2\text{O} \rightarrow \text{e}^- + \text{H}^+ + \frac{1}{2}\text{O}_2$ (water = e^- source)

e^- flow: PSII $\rightarrow$ ETC $\rightarrow$ PSI $\rightarrow$ $\text{NADP}^+ \rightarrow$ NADPH

H^+ pumped INTO thylakoid lumen $\rightarrow$ ATP synthase makes ATP (chemiosmosis)

PRODUCTS: ATP + NADPH + O_2

Stage 2: CALVIN CYCLE (stroma) — "light-independent" but needs ATP/NADPH

CO_2 fixation: rubisco attaches CO_2 to RuBP

Reduction: ATP + NADPH convert to G3P (3-carbon sugar)

Regeneration: G3P recycles to RuBP

NET: 6 $\text{CO}_2 \rightarrow$ 2 G3P $\rightarrow$ eventually 1 glucose

Photosynthesis Variants [4]/[5]

- **[4] C3 plants** (most plants): standard Calvin cycle. Inefficient in hot dry conditions due to **photorespiration** (rubisco binds O_2 instead of CO_2).
- **[5] C4 plants** (corn, sugarcane): separate CO_2 fixation (mesophyll) from Calvin cycle (bundle sheath). Reduces photorespiration. **Adaptation to hot climates.**
- **[5] CAM plants** (cacti, succulents): open stomata at NIGHT (collect CO_2), close during day (run Calvin cycle from stored CO_2). **Adaptation to deserts.**

KEY INSIGHT [4] Both photosynthesis and respiration use chemiosmosis — H^+ gradients power ATP synthase. The mechanism is identical; the source of e^- differs (H_2O for photo; NADH for respiration).

UNIT 4 | CELL COMMUNICATION & CELL CYCLE

Cell Signaling — 3 stages [3]/[4]

STAGE 1: RECEPTION — signal molecule (ligand) binds receptor

Receptor types:

- G-protein coupled receptors (GPCRs) — common, activate G-protein when ligand binds
- Receptor tyrosine kinases (RTKs) — ligand binding triggers phosphorylation cascade
- Ion channel receptors — opens to let ions through

STAGE 2: TRANSDUCTION — signal converted/amplified inside cell

Phosphorylation cascades (kinases activate other kinases)

Second messengers: cAMP, Ca^{2+} , IP_3

STAGE 3: RESPONSE — change in cell behavior

Gene expression, enzyme activity, ion flow, cell movement, cell division

Types of Signaling [4]

- **Direct contact / juxtacrine** — cells touch each other (e.g., immune system).
- **Paracrine** — local signal, neighboring cells (e.g., growth factors).
- **Autocrine** — cell signals itself.
- **Synaptic** — neurons, very short distance, very fast.
- **Endocrine / hormonal** — long distance through bloodstream (e.g., insulin).

Feedback Loops [3]

- **Negative feedback** — response decreases the original stimulus. **Maintains homeostasis.** Examples: thermoregulation, blood glucose (insulin/glucagon), blood pressure.
- **Positive feedback** — response amplifies the stimulus. **Used for irreversible events.** Examples: childbirth (oxytocin), blood clotting, ovulation.

The Cell Cycle [3]

INTERPHASE (cell grows, prepares to divide)

G_1 — cell grows, normal function

S — DNA replication (chromatin → 2 sister chromatids)

G_2 — final growth + checks before mitosis

MITOTIC PHASE

Mitosis (PMAT):

Prophase — chromosomes condense, nuclear envelope breaks down, spindle forms

Metaphase — chromosomes line up at metaphase plate

Anaphase — sister chromatids separate, pulled to poles

Telophase — chromosomes decondense, nuclear envelope reforms

Cytokinesis — cytoplasm divides (cleavage furrow in animals, cell plate in plants)

Cell Cycle Checkpoints [4]

- **G_1 checkpoint** — "restriction point." Cell decides if conditions are favorable for division.
- **G_2 checkpoint** — checks DNA replication completed correctly.
- **M (spindle assembly) checkpoint** — checks chromosomes properly attached to spindle before anaphase.
- **Cyclins + Cdks** — proteins that drive cell cycle progression. Cyclin levels rise/fall in cycle; Cdks must bind cyclins to activate.

Mitosis vs Meiosis [3]/[4]

Feature	Mitosis	Meiosis
Purpose	Growth, repair, asexual reproduction	Gamete (sperm/egg) production
# divisions	1	2 (Meiosis I + Meiosis II)
# daughter cells	2	4
Daughter cell type	Diploid ($2n$), genetically IDENTICAL to parent	Haploid (n), genetically UNIQUE
Genetic variation	None (mutations excluded)	Crossing over (Prophase I) + independent assortment + random fertilization
Where	Somatic (body) cells	Germ-line cells (testes, ovaries)

KEY INSIGHT [5] Cancer = uncontrolled cell division caused by failed checkpoints. **Proto-oncogenes** (when mutated → oncogenes) drive division. **Tumor suppressors** (e.g., p53) normally block division when DNA damaged. Loss of both classes → cancer.

UNIT 5 | HEREDITY · Mendel, Punnett squares, sex-linkage

Mendel's Laws [3]

- **Law of Segregation** — alleles separate during gamete formation; each gamete receives one allele per gene. (Anaphase I)
- **Law of Independent Assortment** — alleles for different genes assort independently if on different chromosomes (or far apart on same).
- **Law of Dominance** — dominant allele masks recessive in heterozygotes.

Punnett Squares — single + dihybrid [3]/[4]

Cross Aa × Aa: 1 AA : 2 Aa : 1 aa (3:1 dominant:recessive ratio).

Dihybrid cross AaBb × AaBb (assuming independent): 9:3:3:1 phenotypic ratio.

Test cross — cross unknown phenotype with homozygous recessive (aa) to determine genotype. If any recessive offspring, parent was Aa.

Beyond Mendel — Non-Mendelian Inheritance [4]/[5]

- **[4] Codominance** — both alleles fully expressed in heterozygote. Example: AB blood type.
- **[4] Incomplete dominance** — heterozygote shows blended phenotype. Example: red × white snapdragon = pink.
- **[4] Multiple alleles** — more than 2 alleles in population (e.g., ABO blood: I^A, I^B, i).
- **[4] Polygenic inheritance** — multiple genes affect single trait (skin color, height). Continuous variation.
- **[5] Pleiotropy** — single gene affects multiple traits (e.g., sickle cell anemia).
- **[5] Epistasis** — one gene's expression depends on another (e.g., labrador coat color: gene B for pigment, gene E for deposition).
- **[5] Linked genes** — genes on same chromosome don't assort independently. Recombination frequency = map distance (cM).

Sex-Linked Inheritance [4]

X-linked recessive disorders (hemophilia, color blindness) affect males more often because males (XY) need only one recessive allele to express the disorder, while females (XX) need two.

X-linked recessive cross: Carrier mother (X^A X^a) × Normal father (X^A Y)

Punnett:

		X ^A	Y	
X ^A	X ^A X ^A	X ^A Y		← 1/4 normal daughters; 1/4 normal sons
X ^a	X ^A X ^a	X ^a Y		← 1/4 carrier daughters; 1/4 affected SONS

Result: 1/2 of sons affected. No daughters affected (could be carriers).

Pedigrees — reading family trees [4]

- **Squares = males, circles = females.** Filled = affected. Horizontal line = mating; vertical = offspring.
- **Autosomal recessive** — skips generations; affected children of unaffected parents (both carriers); affects males and females equally.
- **Autosomal dominant** — every generation affected; affected child has at least one affected parent.
- **X-linked recessive** — mostly males affected; mother of affected son is carrier.

Probability Rules in Genetics [3]/[4]

Multiplication rule (independent events): $P(A \text{ AND } B) = P(A) \times P(B)$
 e.g., Probability of two recessive offspring in a row = $(1/4)^2 = 1/16$

Addition rule (mutually exclusive): $P(A \text{ OR } B) = P(A) + P(B)$
 e.g., Probability of homozygous offspring (AA OR aa) = $1/4 + 1/4 = 1/2$

UNIT 6 | GENE EXPRESSION & REGULATION

The Central Dogma — DNA → RNA → Protein [3]

DNA —TRANSCRIPTION→ mRNA —TRANSLATION→ Protein

In eukaryotes: transcription in NUCLEUS, translation in CYTOPLASM (at ribosomes)

In prokaryotes: both occur in cytoplasm SIMULTANEOUSLY (no nucleus)

DNA Replication [3]

- **Semi-conservative** — each new dsDNA has 1 old + 1 new strand.
- **Helicase** unwinds DNA. **Primase** lays RNA primer. **DNA polymerase III** synthesizes new strand 5'→3' (reads template 3'→5').
- **Leading strand** synthesized continuously toward replication fork. **Lagging strand** synthesized in **Okazaki fragments**, joined by **DNA ligase**.
- **[4] Telomeres** — repetitive DNA at chromosome ends. Shorten with each division. **Telomerase** extends them in stem cells/cancer.

Transcription [3]

- **RNA polymerase** binds promoter (TATA box in eukaryotes). Reads DNA template 3'→5', synthesizes RNA 5'→3'.
- **Eukaryotic mRNA processing** before leaving nucleus:
 - **5' cap** (modified G) added — protects mRNA, helps ribosome bind.
 - **3' poly-A tail** — protects from degradation.
 - **Splicing** — introns removed, exons joined by spliceosome.

KEY INSIGHT [4] Alternative splicing — different exon combinations from same gene → different proteins. Why ~21,000 human genes can make ~100,000+ proteins.

Translation [3]

- **Ribosome reads mRNA codons** (3 bases). 64 codons → 20 amino acids (+ start AUG, 3 stop codons).
- **tRNA** brings amino acid to ribosome. tRNA anticodon pairs with mRNA codon.
- **3 sites on ribosome:** A (incoming aa), P (growing chain), E (exit). Peptide bond forms between aa in P and A.
- **Stop codon** → release factor → polypeptide released.

Mutations [3]/[4]

Type	Description + effect
Point: silent	Substitution that doesn't change amino acid (codon redundancy). No effect.
Point: missense	Substitution changes one amino acid. May or may not affect protein function.
Point: nonsense	Substitution creates stop codon → truncated protein. Usually severe.
Frameshift: insertion / deletion	Adds or removes 1-2 bases → shifts reading frame. Almost always severely damaging — every codon downstream is wrong.
Chromosomal	Larger-scale: deletion, duplication, inversion, translocation. Affects many genes.

Gene Regulation in Prokaryotes — Operons [4]

OPERON = cluster of genes regulated together by one promoter + operator

LAC OPERON (inducible — turned ON by lactose):

Default: repressor binds operator → genes OFF

+Lactose: lactose binds repressor → repressor releases operator → genes ON

Logic: only make lactose-digesting enzymes when lactose is present

TRP OPERON (repressible — turned OFF by tryptophan):

Default: repressor inactive → genes ON (making tryptophan)

+Tryptophan: trp binds repressor → repressor binds operator → genes OFF

Logic: stop making tryptophan when there's plenty

Gene Regulation in Eukaryotes [4]/[5]

- **[4] Transcription factors** bind enhancer/promoter regions to activate or repress transcription.
- **[5] Epigenetic modifications** — DNA methylation (silences genes) and histone acetylation (activates by loosening chromatin). Heritable but reversible.
- **[5] microRNA / RNAi** — small RNAs bind mRNA → degrade it or block translation. Post-transcriptional regulation.

UNIT 7 | NATURAL SELECTION & EVOLUTION

Mechanisms of Evolution — what causes allele frequency change [3]/[4]

Mechanism	Effect
Natural selection [3]	Differential survival/reproduction based on heritable traits. Adaptive — drives populations toward higher fitness.
Mutation [3]	Source of NEW alleles. Random, generally rare. Without mutation, no new variation.
Gene flow [3]	Movement of alleles between populations (migration). Reduces differences between populations.
Genetic drift [4]	Random changes in allele frequencies, especially in small populations. Bottleneck effect (population crashes); founder effect (small group establishes new population).
Non-random mating [4]	Sexual selection, assortative mating. Changes genotype frequencies but not always allele frequencies.

Hardy-Weinberg Equilibrium — the null hypothesis for evolution [4]/[5]

H-W EQUATIONS:

$$p + q = 1 \quad (\text{allele frequencies sum to } 1)$$

$$p^2 + 2pq + q^2 = 1 \quad (\text{genotype frequencies sum to } 1)$$

WHERE:

$$p^2 = \text{frequency of homozygous DOMINANT genotype}$$

$$2pq = \text{frequency of HETEROZYGOUS genotype}$$

$$q^2 = \text{frequency of homozygous RECESSIVE genotype}$$

H-W ASSUMPTIONS (population in equilibrium = NOT evolving):

1. Large population size
2. No mutation
3. No migration / gene flow
4. Random mating
5. No natural selection

[4] **Use H-W to detect evolution.** If observed frequencies differ from H-W predictions, the population IS evolving — one or more assumptions is violated.

[5] **Worked example.** In a population, 16% of individuals show recessive trait. Find allele freqs.

Solution: $q^2 = 0.16 \rightarrow q = 0.4 \rightarrow p = 0.6$. Heterozygous freq = $2pq = 2(0.6)(0.4) = 48\%$. Homozygous dominant = $p^2 = 36\%$.

Types of Natural Selection [4]

- **Directional** — favors one extreme (e.g., antibiotic resistance — favors resistant bacteria).
- **Stabilizing** — favors average phenotype (e.g., human birth weight).
- **Disruptive / diversifying** — favors both extremes (e.g., African seedcrackers with two beak sizes for different seed types).

Speciation [4]/[5]

- **Species** = group of organisms that can interbreed and produce fertile offspring (biological species concept).
- **Allopatric speciation** [4] — geographic isolation $\rightarrow$ reproductive isolation evolves $\rightarrow$ new species. **Most common.**
- **Sympatric speciation** [5] — speciation without geographic isolation. Can happen via polyploidy (plants), niche specialization, sexual selection.
- **Reproductive isolation mechanisms:** prezygotic (habitat, temporal, behavioral, mechanical, gametic) and postzygotic (zygote mortality, hybrid sterility).

Evidence for Evolution [3]/[4]

- **Fossil record** — shows progression of forms over time.
- **Anatomical homologies** — similar structures in different species (whale flipper, human hand) suggest common ancestor.
- [4] **Vestigial structures** — reduced/non-functional features (whale pelvis, human appendix).
- [3] **DNA + protein sequence comparison** — closer species share more sequence.
- [4] **Embryological similarities** — vertebrate embryos look very similar early in development.
- [3] **Direct observation** — antibiotic resistance, peppered moths, Galápagos finches.

Phylogenetic Trees [4]

- **Read like family trees.** Root = common ancestor. Nodes = speciation events. Tips = current species.
- **Closer branch points = more recent common ancestor.** Branch length sometimes represents time.
- [5] **Cladogram vs phylogram:** cladograms show only relationships; phylograms include branch lengths showing genetic divergence.

UNIT 8 | ECOLOGY · populations, communities, ecosystems

Levels of Ecology [3]

- **Organism** → **Population** (same species, same area) → **Community** (multiple populations) → **Ecosystem** (community + abiotic factors) → **Biome** (regional climate).

Population Ecology [3]/[4]

EXPONENTIAL growth (J-curve): $dN/dt = rN$

r = per-capita growth rate. Unlimited resources. Rare in nature.

LOGISTIC growth (S-curve): $dN/dt = rN \cdot (K - N)/K$

K = carrying capacity. Growth slows as N approaches K .

Population stabilizes at K when births = deaths.

r -selected species: many offspring, little parental care, short life. (mosquitos, weeds)

K -selected species: few offspring, much parental care, long life. (whales, humans)

Community Interactions [3]

Interaction	Effect on each species + example
Competition (– / –)	Both harmed. Shared resource. Competitive exclusion: two species can't occupy same niche.
Predation (+ / –)	Predator benefits, prey harmed. Drives prey defenses (camouflage, mimicry, toxins).
Herbivory (+ / –)	Herbivore eats plant. Plants evolve defenses (thorns, chemicals).
Parasitism (+ / –)	Parasite benefits, host harmed (but usually not killed).
Mutualism (+ / +)	Both benefit. (bees + flowers, gut bacteria + humans, mycorrhizae + plants)
Commensalism (+ / 0)	One benefits, other unaffected. (barnacles on whale)

Energy Flow + Trophic Levels [3]

Producers (autotrophs) → Primary consumers (herbivores) → Secondary → Tertiary

~10% RULE: only ~10% of energy passes from one trophic level to the next.

90% lost as heat (respiration), waste, undigested material.

IMPLICATION: shorter food chains support more biomass at top.

Apex predators are rare because energy runs out.

Biogeochemical Cycles [3]/[4]

- [3] **Water cycle** — evaporation, condensation, precipitation, runoff. No major reservoir change.
- [3] **Carbon cycle** — photosynthesis ($\text{CO}_2 \rightarrow \text{glucose}$) + respiration ($\text{glucose} \rightarrow \text{CO}_2$). **Combustion of fossil fuels** adds CO_2 to atmosphere → global warming.
- [4] **Nitrogen cycle** — atmospheric $\text{N}_2 \rightarrow$ fixed by bacteria (legume root nodules) → assimilated by plants → consumed → decomposed → denitrified back to N_2 .
- [4] **Phosphorus cycle** — no atmospheric phase. Weathering of rocks → soil → plants → consumers → decomposers. Often the **limiting nutrient** in ecosystems.

Ecological Disruptions [3]/[4]

- [3] **Climate change** — increased atmospheric CO_2 from fossil fuel combustion → global warming → ecosystem shifts.
- [3] **Biodiversity loss** — habitat destruction, pollution, invasive species, overexploitation, climate change. Reduces ecosystem resilience.
- [4] **Invasive species** — non-native species that disrupt local ecosystems. Often outcompete natives because they lack predators/parasites in new environment.
- [4] **Eutrophication** — nutrient pollution → algal blooms → algae die → bacteria decompose → O_2 depleted → fish die. Common from fertilizer runoff.
- [5] **Keystone species** — species whose removal disproportionately affects ecosystem (e.g., sea otters control sea urchins → kelp forests survive).

PAGE 11 | FRQ STRATEGY · the 6 FRQ types + experimental design

AP Bio FRQ Structure [3]

AP Bio has 8 FRQs total: 2 long, 4 short, plus 2 questions on a single set of data. Time: 90 min for the FRQ section.

Type	Worth	What it asks
Long Q1: Interpreting + Evaluating Experimental Results	8-10 pts	Read graph/data, draw conclusions, propose new experiments
Long Q2: Interpreting + Evaluating Experimental Results with Graphing	8-10 pts	Calculate values, construct a graph, evaluate
Short Q3: Scientific Investigation	4 pts	Design / interpret an experiment
Short Q4: Conceptual Analysis	4 pts	Apply concepts to a scenario
Short Q5: Analyzing Data	4 pts	Read + interpret data
Short Q6: Analyzing a Model	4 pts	Interpret a diagram or model

FRQ DISCIPLINE — non-negotiable for [4]/[5]

- **[3] Use complete sentences.** Bullet points are OK only when prompt says "list" or "identify."
- **[3] Underline or highlight task verbs** in the prompt: "explain," "describe," "identify," "calculate," "justify." Each verb has a different rubric expectation.
- **[3] Answer EVERY part.** Even a partial answer can earn points. Empty parts earn 0.
- **[4] Connect to mechanism.** Don't say "the protein binds and works." Say "the enzyme's active site binds the substrate via H-bonding, lowering activation energy by stabilizing the transition state."
- **[5] Connect across units / Big Ideas.** "This relates to EVO because populations with antibiotic resistance show natural selection in real time." Naming the Big Idea earns sophistication credit.

EXPERIMENTAL DESIGN — the score-defining FRQ skill [4]/[5]

At least one FRQ asks you to design an experiment. The rubric scores these specific elements:

REQUIRED ELEMENTS for full credit:

1. State a HYPOTHESIS — predicts a relationship between variables
2. Identify INDEPENDENT variable (what you change)
3. Identify DEPENDENT variable (what you measure)
4. Identify CONTROL group (or at least one CONTROLLED variable held constant)
5. Describe how you will MEASURE the dependent variable (be specific)
6. Describe sample size, replication, controls
7. Predict the EXPECTED RESULTS if hypothesis is true
8. [5] Identify potential SOURCES OF ERROR or how to address them

DATA ANALYSIS / GRAPHING [4]

- **Always label both axes** with quantity AND units.
- **Identify variables before answering.** Independent variable on x-axis, dependent on y-axis (convention).
- **Use error bars** when given. Overlapping error bars = no significant difference. Non-overlapping = significant difference.
- **[5] Statistical reasoning:** when $p < 0.05$, results are statistically significant. Larger sample size → more reliable conclusions.

AP BIO MATH SKILLS [3]/[4]

Probability: (covered in heredity)
 Hardy-Weinberg: $p^2 + 2pq + q^2 = 1$
 Population growth: $dN/dt = rN$ (exponential) or $rN(K-N)/K$ (logistic)
 Gibbs free energy: $\Delta G = \Delta H - T\Delta S$ (favorable when $\Delta G < 0$)
 Q_{10} (temperature): rate doubles for every 10°C increase (in physiologic range)
 Surface area:Volume: smaller cells = higher SA:V → faster diffusion

PAGE 12 | THE 3/4/5 SCORE LADDER · tier-specific test-day strategy

THE SCORE LADDER — same prompt, three different answers

The clearest way to see what each tier demands. Find your tier in the column on the right and write at that level.

3-scorer writes...	4-scorer writes...	5-scorer writes...
"Cells need O ₂ ."	"Cells use O ₂ in cellular respiration to make ATP."	"O ₂ is the FINAL electron acceptor in the ETC, allowing protons to keep flowing through ATP synthase. Without O ₂ , NADH accumulates and ETC stalls — cells must switch to fermentation, which yields only 2 ATP/glucose vs ~30 with O ₂ ."
"DNA copies itself."	"DNA replication is semi-conservative; helicase unwinds, polymerase adds nucleotides 5'→3'."	"Semi-conservative replication: helicase unwinds the helix; primase lays an RNA primer; DNA pol III synthesizes 5'→3' on each template (3'→5' read direction). Leading strand is continuous; lagging strand requires Okazaki fragments joined by ligase. This connects to IST (Information Storage) because the mechanism preserves genetic information across generations."
"Plants make food from sunlight."	"Photosynthesis converts CO ₂ + H ₂ O to glucose using light energy. Light reactions in thylakoids; Calvin cycle in stroma."	"Light reactions split H ₂ O (photolysis) → e ⁻ flow through PSII → ETC → PSI → NADP ⁺ . H ⁺ pumped into thylakoid lumen; chemiosmosis through ATP synthase. NADPH + ATP feed Calvin cycle, where rubisco fixes CO ₂ to RuBP. This relates to ENE (Energetics) as energy is converted from photons to chemical energy in glucose."
"Mutations cause cancer."	"Mutations in proto-oncogenes (drive division) or tumor suppressors (stop division) cause uncontrolled cell division."	"Cancer arises from accumulating mutations that bypass checkpoints. Proto-oncogenes (e.g., RAS) become oncogenes when mutated, driving constitutive division. Tumor suppressors (e.g., p53) normally trigger apoptosis or G1 arrest in response to DNA damage; when both copies are lost, damaged cells continue dividing. Most cancers require multiple hits."
Identifies the IV and DV in an experiment	Identifies IV, DV, control group, AND describes how to measure DV	Adds sample size + replication + sources of error + alternative explanations

TIER-SPECIFIC TEST-DAY CHECKLISTS

For a 3 — bag ALL of these:

- ✓ MCQ: 32+ correct. Use elimination on the unfamiliar ones; don't leave anything blank.
- ✓ Each FRQ: complete every part with at least an attempt — even partial answers earn points.
- ✓ For experimental questions: ALWAYS state IV, DV, and at least one control.
- ✓ Use complete sentences (unless asked to list).
- ✓ Memorize the cellular respiration + photosynthesis equations.

For a 4 — add these on top:

- ✓ MCQ: 38+ correct. Practice data-based questions.
- ✓ Each FRQ: structured with topic-evidence-explanation. Connect concepts across units.
- ✓ Always describe MECHANISM, not just observation: "the protein folds because hydrophobic R-groups cluster away from water" not just "the protein folds."
- ✓ For graphs: state pattern AND propose explanation. Reference data points specifically.
- ✓ Hardy-Weinberg calculations: practice until automatic.

For a 5 — add these on top:

- ✓ MCQ: 46+ correct. Quick + accurate.
- ✓ Every FRQ: name the relevant **Big Idea** (EVO, ENE, IST, SYI) and tie answer back.
- ✓ Multi-unit synthesis: "This connects to evolution because..." / "At the molecular level, this means..."
- ✓ Discuss limitations + alternative explanations + sources of error in experimental questions.
- ✓ Use specific quantitative data when given. "In Treatment 2, the rate doubled compared to control (12 vs 6 units)."
- ✓ Distinguish correlation from causation. Acknowledge when data is suggestive but not definitive.

THE ONE THING THAT MATTERS MOST [3][4][5]: ANSWER EVERY PART. AP Bio FRQs are scored holistically per part. A blank part earns 0. A poorly-attempted part can earn 1-2 points. Even if you don't fully know the answer, write down what you DO know connecting to the question. **Partial credit adds up across 8 FRQs.**

If you do nothing else: pick your tier, study to that tier, and answer every FRQ part. Match your effort to your target — that's how you get the score.

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